Report on Practical 6

Dynamic Routing

1. Aims

- Implement a dynamic routing setup using Routing Information Protocol (RIP) in Cisco Packet Tracer.

2. Objectives

- Configure and implement dynamic routing using RIP in a network topology.
- Understand the difference between static and dynamic routing.
- Identify scenarios where static or dynamic routing is appropriate.
- Evaluate the advantages and disadvantages of each routing method.

3. Background

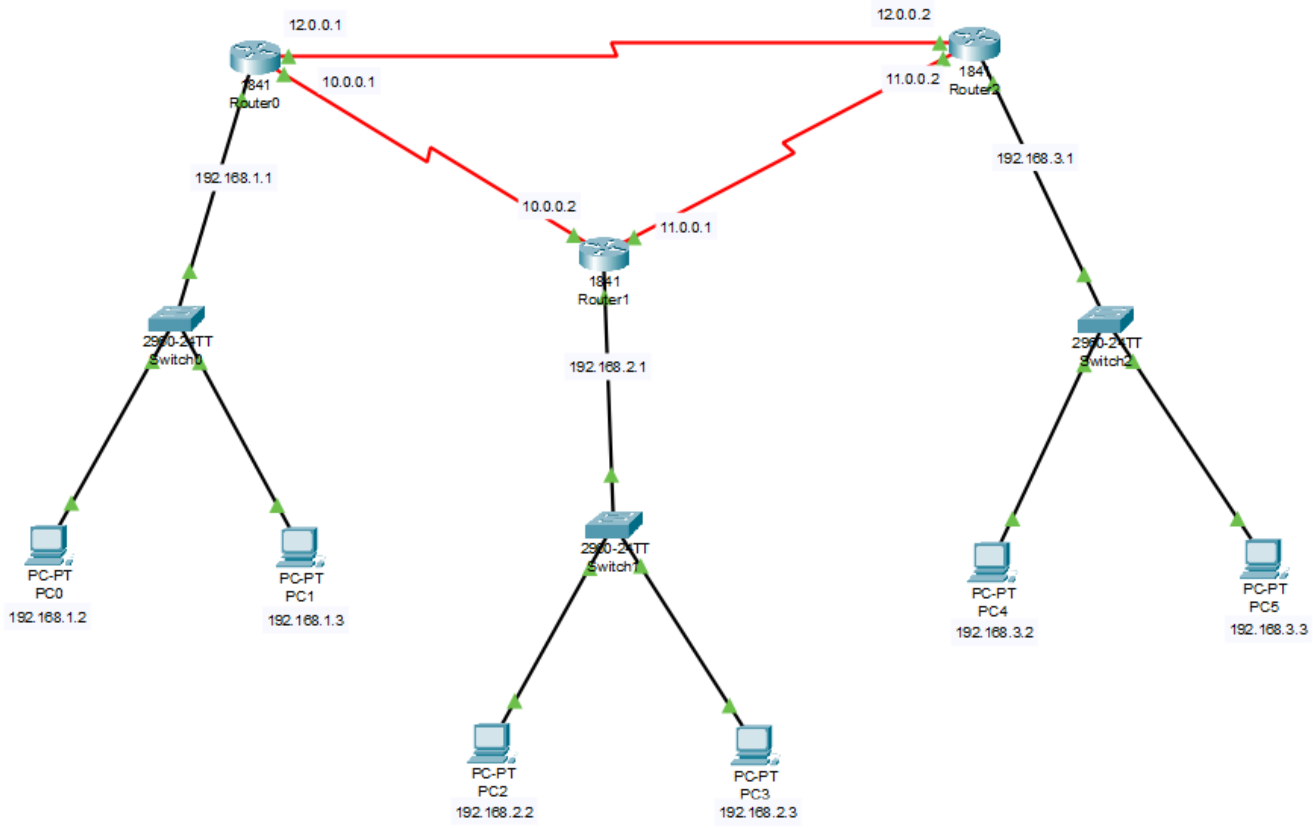
- **Routing** involves selecting paths in a network along which data packets travel and ensures that packets are efficiently delivered across networks
- Routing has two primary types of routing method, which are:

    Static routing

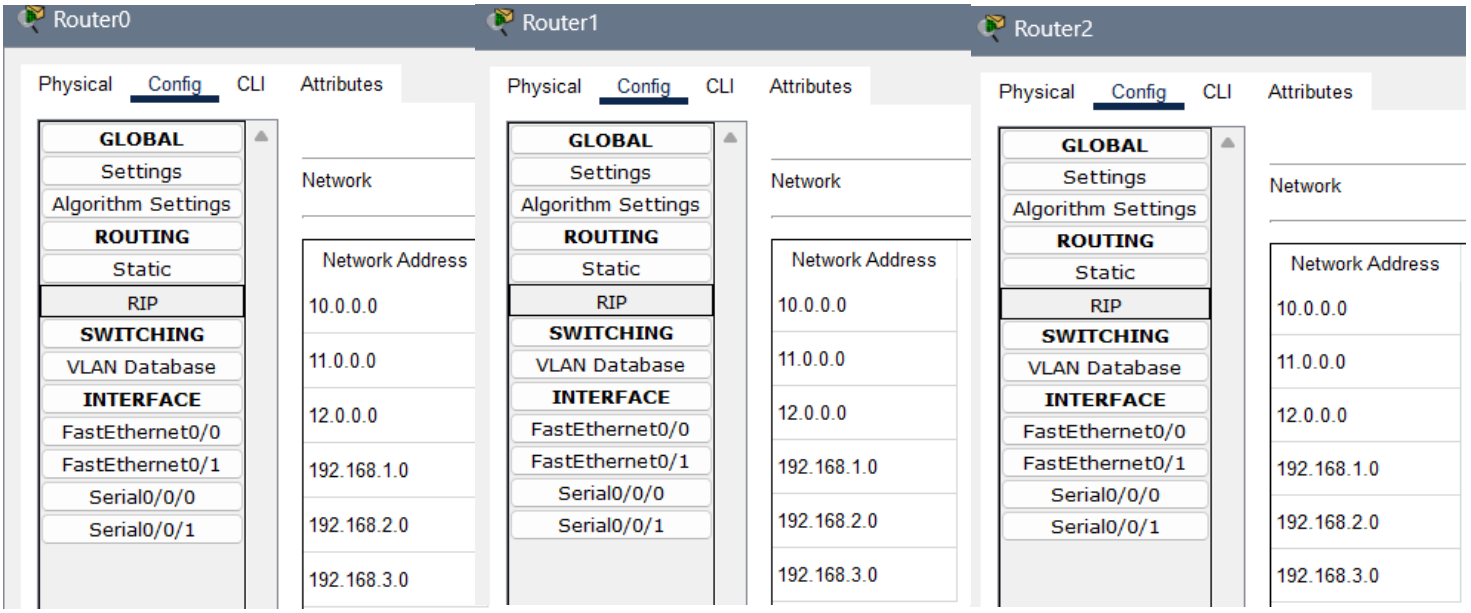
    Dynamic routing
- Dynamic routing is a method where routers automatically exchange information using routing protocols to learn and adapt to changes in the network.
- Dynamic routing is mostly used in larger networks where manual route configuration would be impractical and time-consuming.
- Few protocols of Dynamic routing are RIP, OSFP, EIGRP.
- **RIP** (Routing Information Protocol) is a distance-vector routing protocol that uses hop count as a metric to determine the best path between networks.
- **OSPF** (Open Shortest Path First) is defined as a link-state routing protocol to find the optimum path between the source router and the destination router.
- **EIGRP** (Enhanced Interior Gateway Routing Protocol) is defined as a distance-vector protocol to automate routing configuration and decisions on a network.

4. Procedure

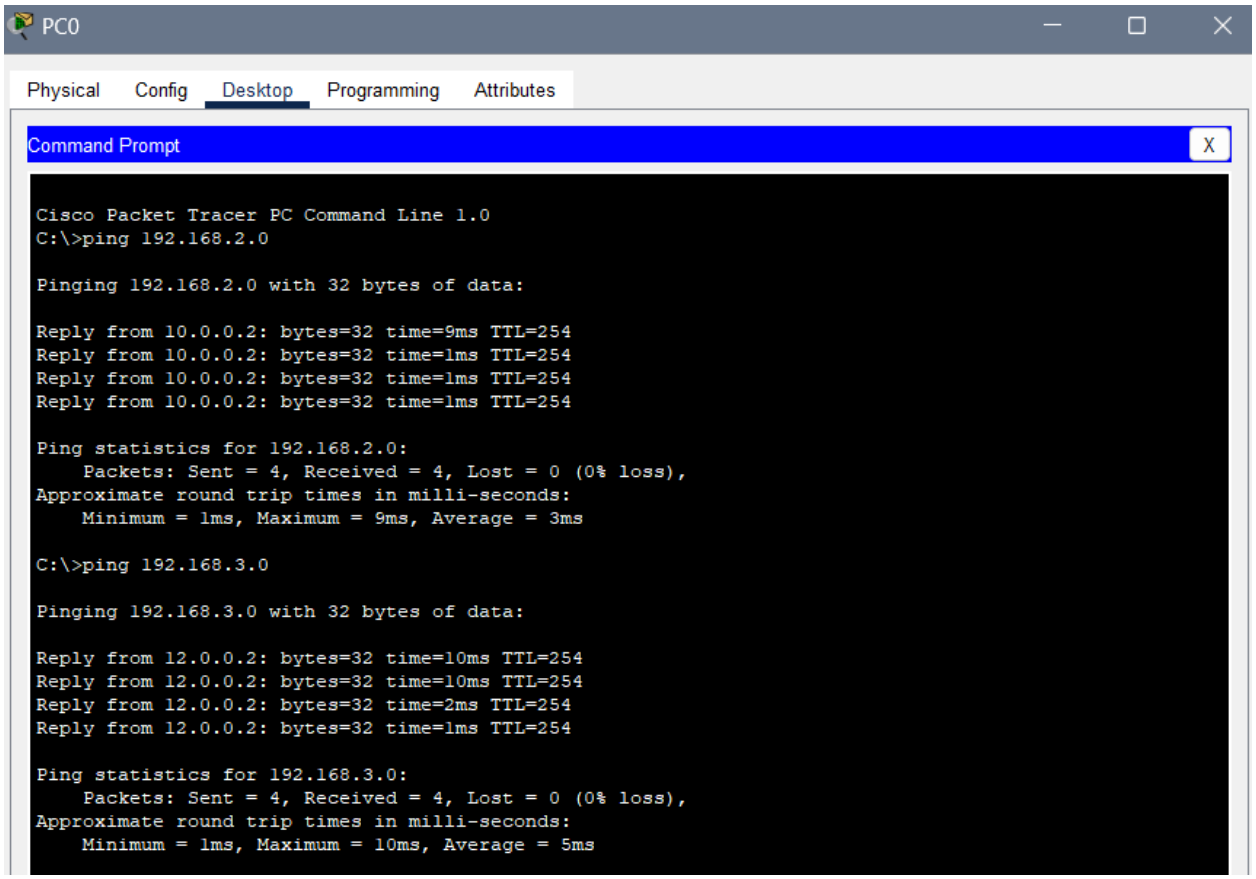
1. Set up a network with multiple routers and PCs as shown in the tutorial and assign IP address.



2. Configure RIP on routers



3. Use the **ping** command to check connectivity.



4. Question And Answer

Q1. Why is dynamic routing different than static routing? What is the difference in implementation between the two.

Dynamic Routing	Static Routing
- Routes get automatically configured by the network administrator. - Routing protocols like RIP, OSPF, or EIGRP automatically discover and maintain routes.	- Routes are manually configured by the network administrator. - They remain fixed unless manually changed.

Q2. Give examples for when dynamic and static routing should be used. Elaborate in detail the pros and cons of each routing method.

Static Routing	Dynamic Routing
Used Cases - Small networks with limited routes. - Networks with predictable and stable traffic. - Situations where network security requires tight control of routes. **Pros** - Simple and predictability. - Low overhead as it doesn't consume bandwidth for route update - Greater control over routes **Cons** - Not scalable for large networks. - High administrative burden for adding/changing routes. - No automatic adaptation to topology changes.	**Used Cases** - Large /complex networks with frequent topology changes. - Networks requiring automatic recovery from link failures. **Pros** - Scalable for large networks. - Automatically adapts to network changes. - Reduces administrative effort. **Cons** - Consumes bandwidth for routing updates. - Requires additional resources (e.g., CPU and memory on routers). - May be harder to troubleshoot compared to static routing.

5. Conclusion

Dynamic routing implemented using RIP is more efficient and scalable than static routing for large, complex networks. However, static routing is simpler and better suited for small, stable networks. In static routing we are required to give network, mask and the next hop, whereas in dynamic we only need to give the network and it automatically finds the next hop, the best path on its own. A good network design often incorporates a combination of both methods to balance performance, simplicity, and resilience.

6. References

- YouTube video: Dynamic routing | RIP version 1 (Routing information protocol)

  https://www.youtube.com/watch?v=i83qrFq3BYQ&list=PPSV